
Honors Differential Equations

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I am extremely busy and only completed part of the homework. To save you the trouble of figuring out what I did not do, I did not complete the following:

- 5.3 1b, 1c, 3b, 3c,

Section 5.3 Problem 1

$$\frac{dH}{dt} = \frac{\partial H}{\partial x} \frac{dx}{dt} + \frac{\partial H}{\partial y} \frac{dy}{dt} = (x - x^3)y + y(x^3 - x) = y(x - x^3 + x^3 - x) = 0.$$

Section 5.3 Problem 3

$$\frac{dH}{dt} = \frac{\partial H}{\partial x} \frac{dx}{dt} + \frac{\partial H}{\partial y} \frac{dy}{dt} = \cos y (-x \sin y + 2y) + (-x \sin y + 2y) (-\cos y) = 0.$$

Section 5.3 Problem 15

Supplemental 1

I find it much neater to write things with inner products and norms, rather than messy integrals. Especially when the integrals are in a fraction, because then they get squished down and look really ugly and I kinda hate it.

Note that $\lambda_n = 0$ if and only if $n = 0$. Also note that since $P_n \in L^2[-1, 1]$ for all $n \in \mathbb{N}$, where we have defined L^2 over $\mathbb{R}$, we have the inner product

$$\langle P_n, P_m \rangle = \int_{-1}^1 P_n(t) P_m(t) dt.$$

Let $n \neq m$ be natural numbers. By logic that has been explored previously, for the case of $\lambda_n, \lambda_m \neq 0$ it suffices to show $\lambda_n \langle P_n, P_m \rangle = \lambda_m \langle P_n, P_m \rangle$. The case of $\lambda_n = 0$ must be explored somewhat since we multiply through by λ_n at the beginning of the proof. Since $\lambda_n \neq \lambda_m$, at most one of these λ values can be 0, so without loss of generality let $\lambda_n \neq 0$. If we show that $\lambda_n \langle P_n, P_m \rangle = \lambda_m \langle P_n, P_m \rangle$ for the case of $\lambda_m = 0$, it follows that $\lambda_n \langle P_n, P_m \rangle = 0$, and thus $\langle P_n, P_m \rangle = 0$. Therefore, it suffices to show this is true for the general case.

Note that

$$\frac{d}{dt} \left((1 - t^2) \frac{dP_n}{dt} \right) = (1 - t^2) \frac{d^2 P_n}{dt^2} - 2t \frac{dP_n}{dt}.$$

It follows that

$$\lambda_n P_n(t) = -\frac{d}{dt} \left((1 - t^2) \frac{dP_n}{dt} \right).$$

Now consider the following.

$$\begin{aligned}\lambda_n \langle P_n, P_m \rangle &= \int_{-1}^1 P_n(t) P_m(t) dt \\ &= \int_{-1}^1 \lambda_n P_n(t) P_m(t) dt.\end{aligned}$$

We can plug in our previous equation to obtain

$$\lambda_n \langle P_n, P_m \rangle = - \int_{-1}^1 \frac{d}{dt} \left((1-t^2) \frac{dP_n}{dt} \right) P_m(t) dt.$$

By the method of integration by parts, we have

$$\begin{aligned}- \int_{-1}^1 \frac{d}{dt} \left((1-t^2) \frac{dP_n}{dt} \right) P_m(t) dt &= - \left((1-t^2) \frac{dP_n}{dt} P_m(t) \Big|_{-1}^1 - \int_{-1}^1 (1-t^2) \frac{dP_n}{dt} \frac{dP_m}{dt} dt \right) \\ &= - \left((1-1) \left(\frac{dP_n}{dt} P_m \right) \Big|_{-1}^1 \right) + \int_{-1}^1 (1-t^2) \frac{dP_n}{dt} \frac{dP_m}{dt} dt \\ &= \int_{-1}^1 (1-t^2) \frac{dP_m}{dt} \frac{dP_n}{dt} dt.\end{aligned}$$

Using our previous equation, it follows that

$$\frac{d}{dt} \left((1-t^2) \frac{dP_m}{dt} \right) = -\lambda_m P_m(t).$$

By the method of integration by parts, it follows that

$$\begin{aligned}\int_{-1}^1 (1-t^2) \frac{dP_m}{dt} \frac{dP_n}{dt} dt &= \left((1-t^2) \frac{dP_m}{dt} P_n(t) \right) \Big|_{-1}^1 - \int_{-1}^1 \frac{d}{dt} \left((1-t^2) \frac{dP_m}{dt} \right) \frac{dP_n}{dt} dt \\ &= (1-1) \left(\frac{dP_m}{dt} P_n(t) \right) \Big|_{-1}^1 + \int_{-1}^1 \lambda_m P_m(t) P_n(t) dt \\ &= \lambda_m \int_{-1}^1 P_m(t) P_n(t) dt \\ &= \lambda_m \langle P_m, P_n \rangle \\ &= \lambda_m \langle P_n, P_m \rangle.\end{aligned}$$

Therefore, the Legendre polynomials are mutually orthogonal.

Supplemental 2

Suppose that

$$f(t) = \sum_{n=0}^{\infty} c_n P_n$$

for some $\{c_n\}_{n=0}^{\infty} \subset \mathbb{R}$, and that this series converges for some $t \in \mathbb{R}$. Fix some arbitrary $k \in \mathbb{N}$. Given that the Legendre polynomials are mutually orthogonal, it follows that

$$\begin{aligned}\langle P_k(t), f(t) \rangle &= \left\langle P_k(t), \sum_{n=0}^{\infty} c_n P_n(t) \right\rangle \\ &= \int_{-1}^1 P_k(t) \sum_{n=0}^{\infty} c_n P_n(t) dt \\ &= \sum_{n=0}^{\infty} \int_{-1}^1 P_k(t) c_n P_n(t) dt \\ &= c_k \int_{-1}^1 (P_k(t))^2 dt \\ &= c_k \|P_k(t)\|_2^2,\end{aligned}$$

where $\|\cdot\|_2$ is the L^2 norm. It follows that

$$c_k = \frac{\langle P_k(t), f(t) \rangle}{\|P_k(t)\|_2^2}.$$

Therefore,

$$f(t) = \sum_{n=0}^{\infty} \frac{\langle P_n(t), f(t) \rangle}{\|P_n(t)\|_2^2} P_n(t).$$

One further remark is that Wikipedia says $\|P_n(t)\|_2 = 1$ if we choose $P_n(1) = 1$. It makes sense, but computing the norm was difficult, so all I was able to extract from this is that $a_0, a_1 = 1$ implies $P_0(t), P_1(t) = 1$, so we require this to hold for the set of Legendre polynomials to be orthonormal. I won't assume this fact, since I didn't prove it, but it's an interesting aside.

Supplemental 3

I used Mathematica for this. This is my first time using Mathematica, so I will attach my code. I don't feel like typing out the coefficients so I will only include the graph. The original function is in blue, and the approximation is in orange.

```

1  f[x_] := Exp[x]*Sin[Pi*x]
2
3  a = Integrate[LegendreP[0, x]*f[x], {x, -1, 1}] / Integrate[LegendreP[0, x]*
4      LegendreP[0, x], {x, -1, 1}]
5
6  b = Integrate[LegendreP[1, x]*f[x], {x, -1, 1}] / Integrate[LegendreP[1, x]*
7      LegendreP[1, x], {x, -1, 1}]
8
9  c = Integrate[LegendreP[2, x]*f[x], {x, -1, 1}] / Integrate[LegendreP[2, x]*
10     LegendreP[2, x], {x, -1, 1}]
11
12 d = Integrate[LegendreP[3, x]*f[x], {x, -1, 1}] / Integrate[LegendreP[3, x]*
    LegendreP[3, x], {x, -1, 1}]
13
14 e = Integrate[LegendreP[4, x]*f[x], {x, -1, 1}] / Integrate[LegendreP[4, x]*
    LegendreP[4, x], {x, -1, 1}]

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13
14 g[x_] = a*LegendreP[0, x] + b*LegendreP[1, x] + c*LegendreP[2, x] + d*LegendreP
    [3, x] + e*LegendreP[4, x]
15
16
17
18 Plot[{f[x], g[x]}, {x, -1, 1}]
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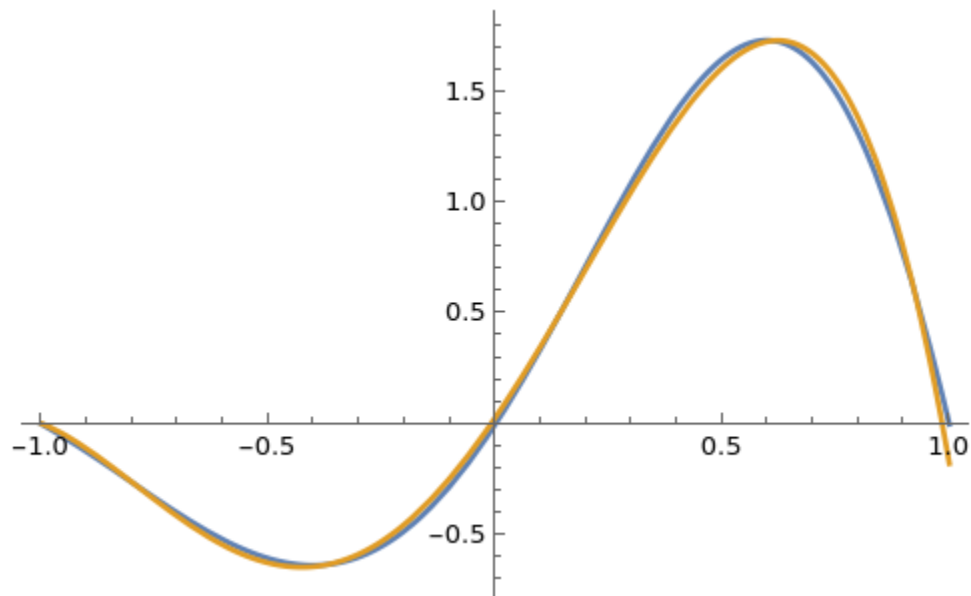


Figure 1: Legendre Approximation